



Inspiring Excellence

CSE370 : Database Systems Project Report

Project Title : BracU Esports Tournament Management System

Group No :16, CSE370 Lab Section :7, Spring 2025		
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22101272	Mashfiq Ferdaus Ahmed	1. Team Records 2. Manage users 3. Profile update
22101086	Md Sakib Hasan Chowdhury	1. Generate fixtures 2. Payment Management

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Introduction

An institution's esports operations or competition can be organized and centralized with the help of the BracU Esports Tournament Management System. The main objective of this system is to facilitate effective management of the students, admins (Those who are in charge of the Tournament) and fun esports game data, while guaranteeing smooth collaboration across the whole BRAC university campus. From managing Player details, Team details, fixtures and details about the video games, this system is designed to create many esports tournaments. So that many esports enthusiasts can join with their team or they can join solo to test their skills and make the tournament enjoyable for all.

Project Features

1. User Management System

- a. User registration and login functionality for Admin and Player.
- b. Profile management for all users, including name, email, password and other details.

2. Role-Based Access Control

- a. Admin can create or modify games, tournaments, fixtures, and game types.
- b. Players can form a group, join a tournament.

3. Registration to Login

- a. Users can directly access to their role-based dashboard without re-login immediately after they register and have access to the features.

4. Payment System

- a. Players/Teams have to give a registration fee for joining a tournament .
- b. Each team has to pay to play in the tournament.

5. Match Fixture Management

- a. Generate a fixture of all teams in a tournament
- b. Insuring every team plays same number of matches with every other team that are in the tournament

6. Team Management

- a. Show payment status of each team.
- b. Show statistics of each team.
- c. Show players in each team.

ER/EER Diagram

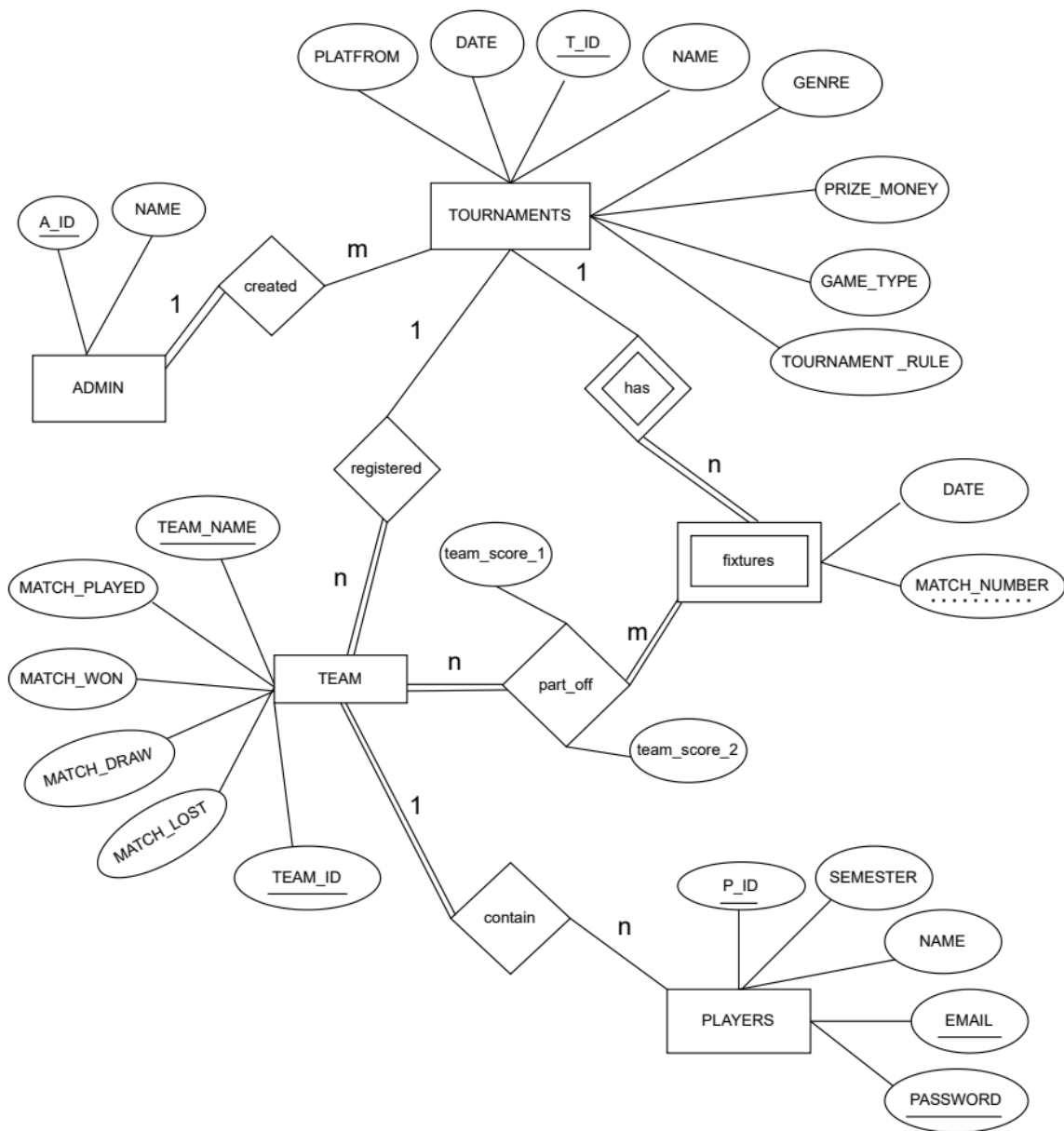
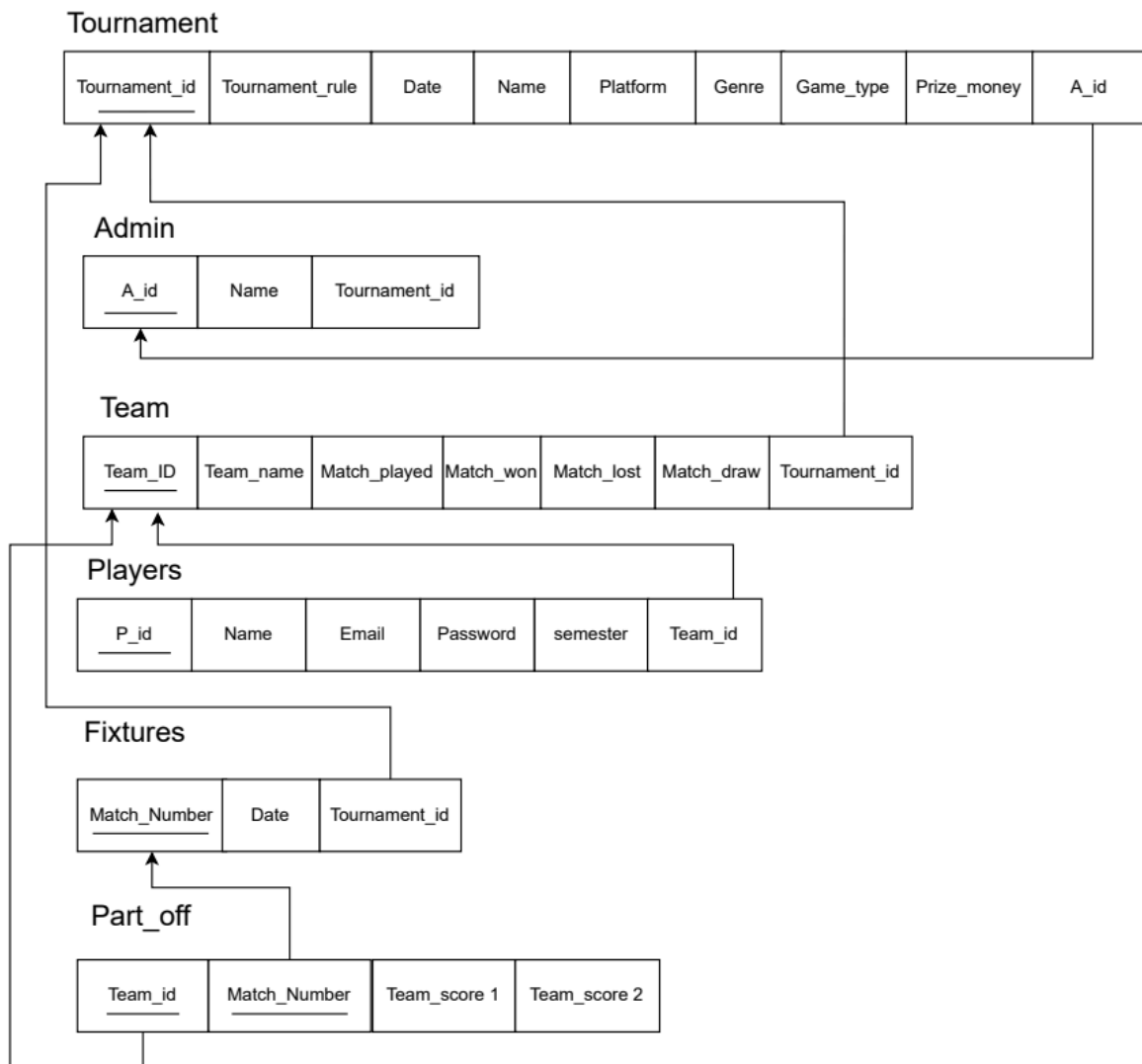


Figure: ER Diagram of “BracU Esports Tournament Management System”

Schema Diagram



Frontend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID :22101811, Name : Abdullah Al Mahamud

1. Admin Authentication System:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Admin Login</title>
  <link href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css" rel="stylesheet">
  <style>
    body {
      background-image: url('../assets/images/all_background.jpg');
      background-size: cover;
      background-position: center;
      background-repeat: no-repeat;
      background-attachment: fixed;
      color: white;
      height: 100vh;
      display: flex;
      align-items: center;
      justify-content: center;
    }
    .login-container {
      background-color: rgba(42, 42, 42, 0.9);
      padding: 30px;
      border-radius: 8px;
      width: 100%;
      max-width: 400px;
      backdrop-filter: blur(5px);
    }
    .form-control {
      background-color: rgba(51, 51, 51, 0.8);
      border: 1px solid #444;
      color: white;
    }
    .form-control:focus {
      background-color: rgba(64, 64, 64, 0.9);
      color: white;
      border-color: #007bff;
    }
    .btn-primary {
      width: 100%;
      background-color: rgba(0, 123, 255, 0.9);
    }
    .error-message {
      color: #dc3545;
      margin-bottom: 15px;
    }
  </style>
</head>
<body>
  <div class="login-container">
    <h2 class="text-center mb-4">Admin Login</h2>
    <?php if($error): ?>
      <div class="error-message text-center"><?php echo $error; ?></div>
    <?php endif; ?>
    <form method="POST" action="">
      <div class="form-group">
        <label for="email">Email</label>
        <input type="email" class="form-control" id="email" name="email" required>
      </div>
    </form>
  </div>
</body>
</html>
```

This frontend focuses on the user interface (UI), enabling the user to interact with the login system.

Key frontend features:

User Input

- Admin login form features email and password fields with required field validation
- Form styled with semi-transparent dark background and backdrop blur
- Input fields use contrasting colors for better visibility
- Error messages displayed in when incorrect credentials are entered

Interactivity

- Login button spans full width with blue background for emphasis
- Form fields change appearance on focus to provide visual feedback
- Client-side validation provides immediate feedback for invalid inputs
- Successful login redirects to admin dashboard automatically

Accessibility and Usability

- Responsive design with max-width for optimal viewing on different devices
- High contrast between text and background colors
- Properly labeled form fields with semantic HTML structure
- Logical tab order for keyboard navigation
- Consistent styling across admin interfaces

Admin Dashboard Navigation

- Buttons for different administrative functions (Add Game, Add Tournament, etc.)
- Font Awesome icons paired with text labels for better recognition
- Color-coded sections to differentiate functional areas
- Active section highlighting to indicate current context
- Consistent logout button placement across all admin pages

2. Add new Games:

```

148 <div class="container">
149 <!-- Predefined Game 1: Valorant -->
150 <section class="game-section">
151 <div class="game-image" style="background-image: url('../assets/images/games-val.jpg');">
152 </div>
153 <div class="game-description">
154 <h2>Valorant</h2>
155 <p>
156 Valorant is a free-to-play first-person tactical hero shooter developed and published by Riot Games. Released in 2020,
157 it quickly rose in popularity within the esports world. Players assume the control of agents, characters with unique
158 abilities, and battle in teams of five. The game combines precise shooting mechanics with strategic, team-based play
159 and objective control.
160 </p>
161 <p>
162 The competitive mode of Valorant features a ranked system that pushes players to climb tiers, adding a thrilling layer
163 of competition. Valorant esports has massive events like the Valorant Champions Tour (VCT) which gathers the best teams
164 from around the world, and offers millions in prize pools.
165 </p>
166 <p>
167 <strong>Key Features:</strong>
168 <ul>
169 <li>5v5 team-based tactical shooting</li>
170 <li>Unique agent abilities</li>
171 <li>Precise gunplay with an emphasis on aim</li>
172 <li>Ranked competitive play</li>
173 </ul>
174 </p>
175 </div>
176 </section>
177 <!-- Predefined Game 2: EA FC24 -->
178 <section class="game-section">
179 <div class="game-image" style="background-image: url('../assets/images/games-fifa.jpg');">
180 </div>
181 <div class="game-description">
182 <h2>EA FC24 (FIFA Series)</h2>
183 <p>
184 EA FC24 (formerly FIFA) is the latest iteration in EA Sports' long-running football simulation series. With EA FC24,
185 players can enjoy a true-to-life soccer experience with real clubs, leagues, and players. The game includes both single-player
186 modes and online multiplayer, with tournaments and leagues dedicated to professional esports players.
187 </p>
188 <p>
189 EA FC24 features iconic modes like Ultimate Team, Career Mode, and Pro Clubs, and introduces new features to enhance
190 gameplay realism. The Esports scene for EA FC24 (previously FIFA) has seen massive tournaments such as the EA Sports FIFA
191 Global Series and the FIFA eWorld Cup, offering players the chance to compete for large prize pools.
192 </p>
193 <p>
194 <strong>Key Features:</strong>
195 <ul>
196 <li>Realistic football simulation</li>
197 <li>Ultimate Team with real players and squads</li>
198 <li>Career Mode and Manager Mode</li>
199 <li>Pro Clubs and online multiplayer</li>
200 </ul>
201 </p>
202 </div>
203 </section>
204

```

```

177 <!-- Predefined Game 2: EA FC24 -->
178 <section class="game-section">
179 <div class="game-image" style="background-image: url('../assets/images/games-fifa.jpg');">
180 </div>
181 <div class="game-description">
182 <h2>EA FC24 (FIFA Series)</h2>
183 <p>
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185 players can enjoy a true-to-life soccer experience with real clubs, leagues, and players. The game includes both single-player
186 modes and online multiplayer, with tournaments and leagues dedicated to professional esports players.
187 </p>
188 <p>
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190 gameplay realism. The Esports scene for EA FC24 (previously FIFA) has seen massive tournaments such as the EA Sports FIFA
191 Global Series and the FIFA eWorld Cup, offering players the chance to compete for large prize pools.
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195 <ul>
196 <li>Realistic football simulation</li>
197 <li>Ultimate Team with real players and squads</li>
198 <li>Career Mode and Manager Mode</li>
199 <li>Pro Clubs and online multiplayer</li>
200 </ul>
201 </p>
202 </div>
203 </section>
204

```

Form Design

- Add Game form includes fields for game name, number of players, description, and game picture
- Form uses standard Bootstrap input controls with clear labels
- File upload field allows administrators to select game images
- Required field validation ensures essential information is provided

User Feedback

- "Add Game" button in admin dashboard reveals the game creation form
- Form submission provides success/error feedback through alert messages
- Successful submission clears the form and displays a confirmation message

Layout and Structure

- Form elements properly labeled for screen readers
- Consistent styling with other admin forms for familiarity
- Form fields arranged in logical order with appropriate spacing
- Error messages clearly indicate issues with form submission
- Game management interface displays existing games in a structured table format

Navigation Elements

- "Add Game" section accessible via dedicated button in admin navigation
- Game management interface includes options to edit and delete existing games
- Form container uses consistent styling with other admin sections
- Game data presented in organized tables with action buttons for management

3. Create Tournament:

```
118 <DOCTYPE html>
119 <html lang="en">
120 <head>
121 <meta charset="UTF-8">
122 <meta name="viewport" content="width=device-width, initial-scale=1">
123 <title>Edit Tournament - Admin Panel</title>
124 <link rel="stylesheet" href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
125 </head>
126 <body>
127 <div class="container">
128 <div class="row">
129 <div class="col-md-3">
130 <div class="list-group">
131 <a href="#">Home</a>
132 <a href="#">Users</a>
133 <a href="#">Games</a>
134 <a href="#">Tournaments</a>
135 <a href="#">Teams</a>
136 <a href="#">Matches</a>
137 <a href="#">Payments</a>
138 <a href="#">Logout</a>
139 </div>
140 </div>
141 <div class="col-md-9">
142 <div class="card">
143 <div class="card-header">
144 <h3>Edit Tournament</h3>
145 </div>
146 <div class="card-body">
147 <div class="form-group">
148 <input type="text" class="form-control" id="tournament_name" value="tournament_name">
149 </div>
150 </div>
151 </div>
152 </div>
153 </body>
154 </html>
```

```
118 <DOCTYPE html>
119 <html lang="en">
120 <head>
121 <meta charset="UTF-8">
122 <meta name="viewport" content="width=device-width, initial-scale=1">
123 <title>Edit Tournament - Admin Panel</title>
124 <link rel="stylesheet" href="https://stackpath.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css">
125 </head>
126 <body>
127 <div class="container">
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129 <div class="col-md-3">
130 <div class="list-group">
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132 <a href="#">Users</a>
133 <a href="#">Games</a>
134 <a href="#">Tournaments</a>
135 <a href="#">Teams</a>
136 <a href="#">Matches</a>
137 <a href="#">Payments</a>
138 <a href="#">Logout</a>
139 </div>
140 </div>
141 <div class="col-md-9">
142 <div class="card">
143 <div class="card-header">
144 <h3>Edit Tournament</h3>
145 </div>
146 <div class="card-body">
147 <div class="form-group">
148 <input type="text" class="form-control" id="tournament_name" value="tournament_name">
149 </div>
150 <div class="form-group">
151 <input type="text" class="form-control" id="description" value="description">
152 </div>
153 <div class="form-group">
154 <input type="text" class="form-control" id="game_type" value="game_type">
155 </div>
156 <div class="form-group">
157 <input type="text" class="form-control" id="tournament_date" value="tournament_date">
158 </div>
159 <div class="form-group">
160 <input type="text" class="form-control" id="entry_fee" value="entry_fee">
161 </div>
162 </div>
163 </div>
164 </div>
165 </body>
166 </html>
```

```

154 <html lang="en">
155 <body>
156 <div class="container">
157 <div class="edit-container">
158 <div class="form-group">
159 <label for="prize_pool">Prize Pool</label>
160 <input type="number" step="0.01" class="form-control" id="prize_pool" name="prize_pool"
161 value="{php echo htmlspecialchars($tournament['prize_pool']); }"
162 placeholder="Enter new prize pool">
163 <div class="current-value">Current: {php echo htmlspecialchars($tournament['prize_pool']); }</div>
164 </div>
165 <div class="form-group">
166 <label for="slots">Available Slots</label>
167 <input type="number" class="form-control" id="slots" name="slots"
168 value="{php echo htmlspecialchars($tournament['slots']); }"
169 placeholder="Enter new number of slots">
170 <div class="current-value">Current: {php echo htmlspecialchars($tournament['slots']); }</div>
171 </div>
172 <div class="form-group">
173 <label for="status">Status</label>
174 <select class="form-control" id="status" name="status">
175 <option value="upcoming">{php echo ($tournament['status'] == 'upcoming') ? 'selected' : '' }</option>
176 <option value="ongoing">{php echo ($tournament['status'] == 'ongoing') ? 'selected' : '' }</option>
177 <option value="completed">{php echo ($tournament['status'] == 'completed') ? 'selected' : '' }</option>
178 <option value="cancelled">{php echo ($tournament['status'] == 'cancelled') ? 'selected' : '' }</option>
179 </select>
180 <div class="current-value">Current: {php echo htmlspecialchars($tournament['status']); }</div>
181 </div>
182 <div class="text-center mt-4">
183 <button type="submit" class="btn btn-primary">Update Tournament</button>
184 <a href="admin.php" class="btn btn-secondary mt-2">Back to Admin Panel</a>
185 </div>
186 </div>
187 </div>
188 </body>
189 </html>

```

Form Design

- Tournament creation form includes comprehensive fields for all tournament details (name, game, fees, dates, slots, banner)
- Game selection implemented as dynamic dropdown menu populated from database
- Form uses Bootstrap styling with responsive layout for cross-device compatibility

User Feedback

- Form submission provides immediate success/error feedback through color-coded alert messages
- Required field validation prevents submission of incomplete tournament information
- Error handling provides specific feedback when tournament creation fails

Layout and Structure

- Form elements arranged in logical sequence with clear visual hierarchy
- Form heading includes Font Awesome trophy icon for visual identification
- Input fields sized appropriately for their content types with consistent spacing

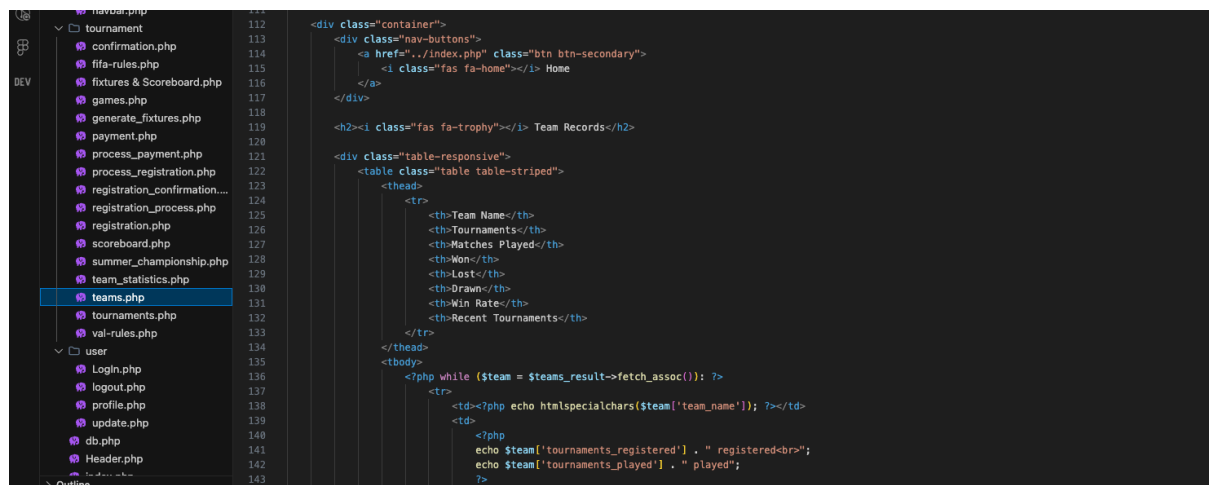
Navigation Elements

- "Add Tournament" section accessible via dedicated button in admin dashboard
- Displays existing tournaments in organized table format

Contribution of ID :22101232, Name : Mashfiq Ferdaus Ahmed

Team :

Screen shot

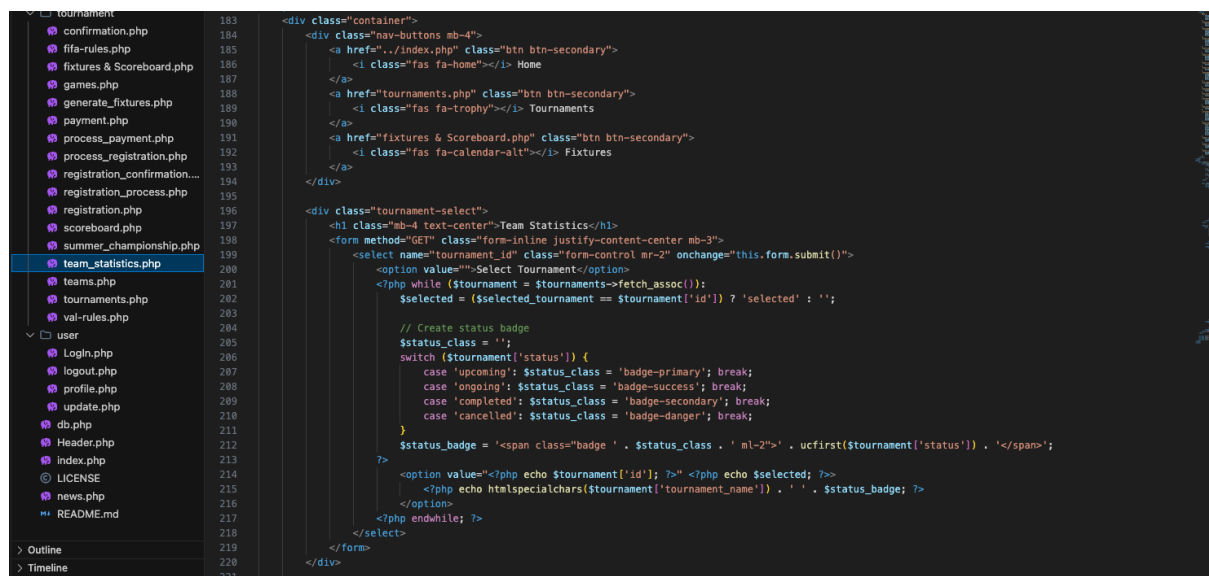


Key Fronted features:

- 1) Fetches team and tournament data using PHP variables
- 2) Dark theme with custom styling , Colored badges for status indicators , Responsive Bootstrap table
- 3) Navigation button to go back to Home

Team Statistics :

Screen shot



Key Fronted features:

- 1) Added dark theme style to the page including background colors, paddings, custom styles, table highlights
- 2) Has buttons for Home , Tournament and Fixture and scoreboard
- 3) When a user selects a tournament it triggers the backend to load data for that tournament. It will show name, game name, number of players, date, status (with colour coded badge)

- 4) Show team statistics table. Here the win rate is calculated and top 3 ranks will be highlighted.
- 5) If no team exists in the selected tournament it will display 'No teams registered' .

User :

Key Fronted features:

- 1) User table display (User id, name,email,phone number)
- 2) An action button to perform
 - a) View = Shows a modal with full user details in a non-editable format
 - b) Edit = Opens a form modal where you can update name, email, and phone. which triggers edit logic in the backend:
 - c) Delete = Asks for confirmation, then submits which triggers edit delete in the backend:
- 3) Shows success or error messages after an action (like update or delete), then resets.

Profile :

Screen shot

```

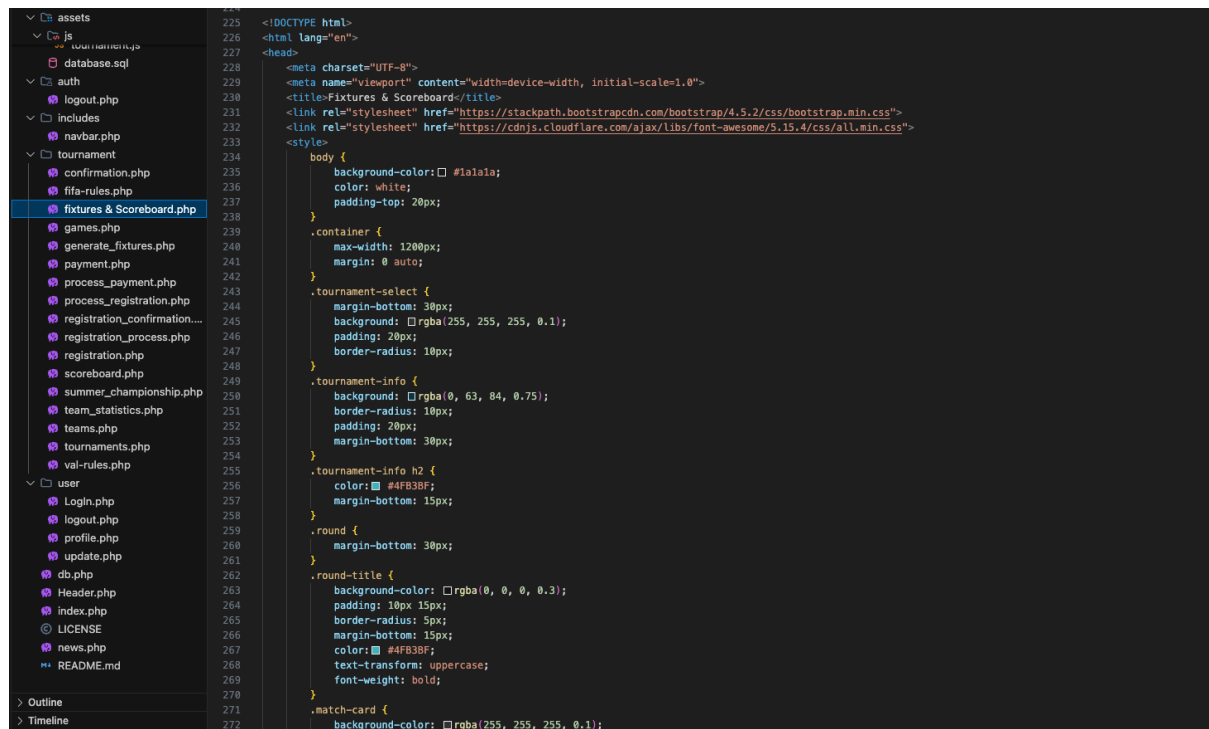
219 <div class="container">
220 <div class="profile-container">
221 <div class="text-center mb-4">
222 <?php if (isset($user['picture']) && $user['picture']): ?>
223 ?php echo htmlspecialchars($user['picture']); ?> alt="Profile Picture" class="profile-picture">
224 <div class="picture-controls">
225 <form method="POST" style="display: inline;">
226 <button type="submit" name="delete picture" class="btn btn-danger btn-sm onclick="return confirm('Are you sure you want to delete
227 <i class="fas fa-trash"></i> Delete Picture
228 </button>
229 </form>
230 </div>
231 <?php else: ?>
232 
233 <?php endif; ?>
234 <h2>User Profile</h2>
235 </div>
236
237 <?php if (isset($success_message)): ?>
238 <div class="alert alert-success"><?php echo htmlspecialchars($success_message); ?></div>
239 <?php endif; ?>
240 <?php if (isset($error_message)): ?>
241 <div class="alert alert-danger"><?php echo htmlspecialchars($error_message); ?></div>
242 <?php endif; ?>
243
244 <form method="POST" action="" enctype="multipart/form-data">
245 <div class="form-group">
246 <label>Name</label>
247 <input type="text" name="name" class="form-control" value="<?php echo isset($user['name']) ? htmlspecialchars($user['name']) : '' ?>" req
248 </div>
249 <div class="form-group">
250 <label>Phone Number</label>
251 <input type="tel" name="number" class="form-control" value="<?php echo isset($user['number']) ? htmlspecialchars($user['number']) : '' ?>"
252 </div>
253 <div class="form-group">
254 <label>Email</label>
255 <input type="email" name="email" class="form-control" value="<?php echo isset($user['email']) ? htmlspecialchars($user['email']) : '' ?>"
256 </div>
257 <div class="form-group">
258 <label>Team Name (Optional)</label>
259 <input type="text" name="team_name" class="form-control" value="<?php echo isset($user['team_name']) ? htmlspecialchars($user['team_name'])
260 </div>
261 <div class="form-group">
262 <label>Profile Picture (Optional)</label>
  
```

Key Fronted features:

- 1) Profile display section where the user has uploaded a profile picture, it displays it otherwise, it shows a default avatar image. If a new picture is added or an old picture is removed it displays feedback after updating or deleting actions.
- 2) Profile edit form where it required a name, number, email address. Optional being team name and profile picture.
- 3) Actions buttons to return to the home page, log out or delete their entire profile.
- 4) Shows a table of past tournament registrations, joining, to get tournament name and to get team name. Retrieves the names of all players on the registered team. Displays player names under the "Players" column. If there is no registration it will display a message.

Contribution of ID :22101086, Name : Md Sakib Hasan Chowdhury

Generate fixtures(1):



Key Frontend features:

1. Displaying the generated fixtures in a clean, organized, and visually appealing manner.
2. Ensuring that input fields (tournament ID) are correctly filled out before sending data to the server.
3. Providing visual feedback to the user (success or error messages) after the form is submitted to indicate if the fixture generation was successful or if there was an issue.

Generate fixtures:(Interaction)

Key Frontend features:

- 1.The frontend will interact with this backend by presenting options for admins to generate fixtures, display messages based on the success or failure of backend operations, and show tournament and fixture details.
- 2.Frontend can send data tournament id via POST requests to the backend for fixture generation.

3. The frontend can display success or error messages based on the session variables set by the backend.

Fixtures and scoreboard:

Key Frontend features:

1. Provide a dropdown to select a tournament from the available list of tournaments. Display tournament details like the number of registered teams and fixture status next to each tournament.

2. Display a list of fixtures, sorted by round and match number. Each match will show the teams playing, their scores, and the match status.

3. Use color-coded indicators (pending, live, completed) to show the match status clearly.

4. Show a button for generating fixtures only when the tournament is ready.

5. For each match, display a card with the match number, teams, score, and status.

6. Show the teams playing, their scores, and a "winner" label for completed matches.

7. Ensure that only admins can see the fixture generation button and access certain pages.

8. Use alert messages to notify admins or users of success or failure in generating fixtures.

Payment:

```
84 <body>
85 <div class="registration-container">
86 <div class="registration-form">
87 <form id="paymentForm" action="process_payment.php" method="POST">
88 </div>
89 </div>
90 <div id="bKash_fields" class="payment-fields" style="display:none">
91 <div class="alert alert-info">
92 <p>strong-bKash Payment Instructions:</strong></p>
93 <ol>
94 <li>Go to your bKash Mobile Menu by dialing *247#</li>
95 <li>Choose "Send Money"</li>
96 <li>Enter the bKash Account Number: @XXXXXXXXXX</li>
97 <li>Enter amount: @<php echo number_format($registration['entry_fee'], 2);></li>
98 <li>Enter your bKash PIN to confirm</li>
99 <li>Copy the Transaction ID and paste it below</li>
100 </ol>
101 </div>
102 <div class="form-group">
103 <label>bKash Number</label>
104 <input type="text" class="form-control" name="bKash_number" placeholder="@XXXXXXXXXX">
105 </div>
106 <div class="form-group">
107 <label>Transaction ID</label>
108 <input type="text" class="form-control" name="bKash_transaction" placeholder="TR1234567890">
109 </div>
110 </div>
111 <div id="Nagad_fields" class="payment-fields" style="display:none">
112 <div class="alert alert-info">
113 <p>strong-Nagad Payment Instructions:</strong></p>
114 <ol>
115 <li>Go to your Nagad Mobile Menu by dialing *167#</li>
116 <li>Choose "Send Money"</li>
117 <li>Enter the Nagad Account Number: @XXXXXXXXXX</li>
118 <li>Enter amount: @<php echo number_format($registration['entry_fee'], 2);></li>
119 <li>Enter your Nagad PIN to confirm</li>
120 <li>Copy the Transaction ID and paste it below</li>
121 </ol>
122 </div>
123 <div class="form-group">
124 <label>Nagad Number</label>
125 <input type="text" class="form-control" name="Nagad_number" placeholder="@XXXXXXXXXX">
126 </div>
127 <div class="form-group">
128 <label>Transaction ID</label>
129 <input type="text" class="form-control" name="Nagad_transaction" placeholder="TR1234567890">
130 </div>
131 </div>
132 <div id="card_fields" class="payment-fields" style="display:none">
133 <div class="form-group">
134 <label>Card Number</label>
135 <input type="text" class="form-control" name="card_number" placeholder="1234 5678 9012 3456">
136 </div>
137 </div>
138 </div>
139 </div>
140 </div>
141 </div>
142 </div>
143 </div>
144 </div>
```

Key Frontend features:

1. The payment options change dynamically when the user selects a different payment method (cash, bKash, Nagad, card).

- 2.The payment instructions change based on the selected payment method (JavaScript listens to the change event on radio buttons).Input fields appear/disappear dynamically based on the selected payment method.
- 3.The form is submitted without page refresh using AJAX to communicate with the back-end which improves the user experience.
- 4.Basic validation (e.g., ensuring that the form fields are completed and that a valid payment method is selected).

Manage Payment:

Key Frontend features:

1. It fetches tournament and registration data from the backend and displays it in a table, showing user-friendly data like registration ID, tournament name, payment status, etc.
- 2.Includes a form that allows the admin to update the payment status for each registration. It uses a dropdown select < > for the admin to select the payment status and submit it for updating.
3. Displays various status indicators (payment status and tournament status) with different colors, making the information visually clear.
- 4.Utilizes Bootstrap for layout and styling, making the page responsive and visually appealing. Features like tooltips, buttons, and form controls improve user interaction.
- 5.Dropdowns and buttons for updating payment statuses are interactive and make the page functional for admins.

Process Payment:

Key Frontend features:

1. Clear and intuitive design, with visually appealing layouts, easy-to-navigate interfaces, and responsive design for mobile and desktop users.
2. Ensuring users provide correct data before sending it to the back end (payment information, registration details).
3. Sending and receiving data from the server without reloading the page, which is part of a smooth user experience.

Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 22101811, Name : Abdullah Al Mahamud

Admin and User Authentication System:

Authentication Logic:

- Session-based authentication using PHP session management
- Admin credentials verified against hardcoded values (email and password)
- Session variables are set upon successful login (admin_email and admin_logged_in)
- Automatic redirection to admin dashboard after successful authentication
- Session destruction on logout with page redirection
- Control the

Security Implementation:

- Session validation on all admin pages prevents unauthorized access
- Conditional redirects for already logged-in administrators
- Error handling for invalid login attempts
- Complete session termination during the logout process
- Protection against direct URL access to admin pages

Database Operations:

- Using hardcoded credentials for authentication
- Session data is stored server-side for security
- Login status is maintained across page requests
- No sensitive data exposed in client-side code
- Error messages sanitized to prevent information leakage

Backend Development: Game Management:

Data Processing:

- Form data validation and sanitization before database operations
- Prepared statements used to prevent SQL injection attacks
- Transaction-based operations for maintaining data integrity
- Error handling with specific error messages for debugging
- Database queries optimized for performance

File Handling:

- Image upload functionality with file type validation
- Directory creation with appropriate permissions if not exists
- Unique file naming to prevent overwriting existing files
- File movement from temporary location to permanent storage
- Default image assignment when no image is uploaded

Database Operations:

- INSERT queries for adding new game records
- Foreign key constraint checking before deletion operations
- Game retrieval for dropdown population in tournament creation
- Transaction rollback on error to maintain database consistency
- Parameterized queries for all database operations

Backend Development: Tournament Management:

Data Processing:

- Comprehensive form data collection with type casting (intval, floatval)
- Game name retrieval from game ID using database query
- Date validation and formatting for tournament scheduling
- Required field validation server-side
- Transaction-based operations for data integrity

File Operations:

- Tournament banner upload with directory creation
- Default image assignment when no banner is uploaded
- File type validation and secure storage
- Proper file path storage in database
- Error handling for file upload failures

Database Integration:

- Multi-table operations across games and tournaments tables
- Foreign key relationships maintained between tables
- Tournament status management (upcoming, ongoing, completed)
- Fixture generation with team randomization algorithm
- Notification system for tournament changes affecting users

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Team :**Key backend features:**

- 1) Fetches team record how many tournaments the team has signed up for (Regardless of payment)
- 2) It will calculate that how many tournaments they've actually participated in (payment status is confirmed)
- 3) Fetches queries like team id, tournament name, payment status, tournament status.
- 4) Organize Registrations by Team ID into an associative array

Team Statistics :**Key backend features:**

- 1) Fetch all tournaments with all registered teams . Here it includes a sub-query that counts how many teams that have confirmed payments.
- 2) See if a specific tournament is selected. If so then it brings all the data
- 3) Fetch all the stats of those registered teams

Get registered team :**Key backend features:**

- 1) Check for tournament id then prepare SQL statement.
- 2) Teams that registered for the specified tournament (While there payment is confirmed)

User :**Key backend features:**

- 1) Firstly check if the admin is logged in using session variables. If not then it takes to admin login page
- 2) Handle user deletion or update. Safely deletes a user based on their id using a prepared statement or incase of an update then it collects the update data from a form (name, email, number) then uses a prepared statement to update the user's info.
- 3) If all is done properly it will show a success message if not then error message and then redirect to user.php.
- 4) Fetches all user records from the user table in descending order of ID (newest first).

Profile :**Key backend features:**

- 1) Fetch currently logged-in user's data from the database. If no user is found (maybe deleted), logs out the session.
- 2) A Delete Profile operation where it deletes related everything that is connected to the team registration and teams, also deletes profile picture if it exists and finally deletes this user from the user database table. After a successful deletion it destroys session and redirects to registration
- 3) Also there is a delete profile picture when it sets the picture column to null and refreshes the data.
- 4) Update profile (Handles name, number, email, team name, picture(optional)). If another picture is uploaded then it deletes the old picture and saves the new.

Update :**Key backend features:**

- 1) Grabs the user id from the session and retrieves the user's data from the user table.
- 2) Handle simple form submission. Updates the user's email and number in the database using the sanitize inputs.

Generate fixtures(1):

Key backend features:

- 1.Checks if the admin is logged, which is a server-side session management feature.
2. Performs various database operations such as fetching tournament details, checking the number of confirmed teams, generating fixtures, and updating records
3. Contains the logic to determine tournament rounds based on the number of teams, validate if the slots are filled, and generate fixtures dynamically. This is all handled server-side.
- 4.The use of database transactions ensures that fixture generation and tournament status updates are done atomically, which is part of back-end logic.
- 5.The form submission is processed on the server to generate the tournament fixtures based on the input data.
- 6.Handling queries, updating records, and ensuring data integrity using transactions.
- 7.The ability to dynamically generate tournament fixtures, validate input data, and determine the rounds based on the number of confirmed teams.

Generate fixtures:

Key backend features:

- 1.Ensures that only an admin can generate fixtures, enhancing security.
- 2.Performs queries to retrieve tournament details, teams, and fixtures
- 3.Based on the number of teams, it determines how many rounds and matches should be created.
4. Uses transactions to ensure that either all database operations succeed or fail together, maintaining data integrity.

Fixtures and scoreboard:

Key backend features:

- 1.Allow admins to create a new tournament by entering its name, date, slots (maximum teams), etc.
- 2.View tournaments with details like name, date, registered teams, fixtures status, etc.
- 3.Automatically update the status of tournaments based on the match progress (upcoming, ongoing, completed).
- 4.Generate fixtures based on the teams registered for the tournament, ensuring at least 4 teams are registered.Fixtures will automatically be ordered by round and match number. Admins can trigger this feature only if no fixtures exist for the tournament.
- 5.Track and update match scores, results, and status (pending, live, completed).

- 6.Ensure that each tournament has at least 4 teams before generating fixtures.
- 7.Implement a check to prevent the generation of fixtures if they already exist.
- 8.Handle cases like lack of permissions, missing teams, or errors during fixture generation.
- 9..Ensure that only admins can generate fixtures.
- 10.Display appropriate error messages when there are issues such as insufficient teams, failed fixture generation, or unauthorized access.

Payment:

Key backend features:

- 1.The system uses a session start function to manage user authentication and ensure that users are logged in before accessing the payment page.
2. Handling payment is delegated to another PHP script where back-end logic will likely handle transactions, update the database, and notify the user about payment status.

Manage Payment:

Key backend features:

1. Ensures that only authorized admins can access this page by verifying the session.Redirects unauthorized users to the login page, maintaining secure access control.
- 2.Ensures that all database operations (updating payment status and tournament status) are executed atomically, preventing data corruption or inconsistency in case of errors.
- 3.Allows admins to update the payment status of a registration (pending,processing,confirmed,rejected)
4. Displays payment status with an appropriate color-coded system for quick recognition.
- 5..Fetches and updates data related to tournament registrations, payment status, and slot availability.
- 6.Logs errors and transaction success/failure, making it easier to debug issues.
7. Contains multiple SQL queries to fetch tournament and registration data, update payment statuses, and change tournament status based on conditions.
8. It uses the begin transaction() function to handle database transactions, which is a backend feature to ensure data integrity and consistency.
9. Error logging is done server-side to log exceptions related to database queries and payment status updates.
- 10.The backend validates incoming data, ensuring proper payment status values and handling registration updates.

Process Payment:
Key backend features:

- 1.It checks if the user is logged in, which is a server-side session management feature.
- 2.The code interacts with a database performing actions such as verifying user registration, processing payment methods, and updating payment details. This is the core functionality of the back-end.
- 3.It includes the logic for verifying registration details, handling different payment methods (cash, bKash, card), and managing payment status. This logic occurs on the server, not the client-side (front end).
- 4.The use of database transactions to ensure data integrity is a back-end feature.
- 5..Managing various payment methods (like card, bKash, etc.), securely handling payment data.
- 6.Providing proper validation for inputs, handling errors, and rolling back transactions in case of failures.
- 7.Protecting user data (such as payment details) and preventing attacks like SQL injection, cross-site scripting.

Source Code Repository

Upload source code to GitHub or Google Drive and share a publicly accessible link in this section of the report.

<https://drive.google.com/file/d/12rLIr5msnUEkidp3SPvd8Ty3wM1a4Bg5/view?usp=sharing>

Conclusion

An organized and thorough method of handling competitive event data is demonstrated by the creation of the "**Esports Tournament Management**" database system, as depicted in the Entity-Relationship (ER) diagram. The system effectively represents the main entities such as players, teams, admins, tournaments and fixtures along with their complex interactions and characteristics.

This study emphasizes the importance of a well-structured database in streamlining the management of esports events. The design ensures data consistency, integrity, and accessibility across a range of operations, including user registration, creation and modifications of tournaments, match scheduling, Formation of a teams and many more.

All things considered, this project demonstrates our capacity to create Esports tournaments and manage individual players or manage a team and turn into an optimal database design, and deliver a solution that effectively meets the real-world needs of esports tournament management. Through collaboration and problem-solving, we laid the foundation for a reliable and efficient database system, so that we can enjoy our favourite esports or support a team we like without any inconvenience.

References

- 1) Bro Code. (2023, March 29). *PHP Full Course for non-haters* 🐼 [Video].
YouTube. <https://www.youtube.com/watch?v=zZ6vybT1HQs>
- 2) GeeksforGeeks. (2024, March 13). *HTML Complete Guide – A to Z HTML Concepts*. GeeksforGeeks. <https://www.geeksforgeeks.org/html-complete-guide/>
- 3) W3Schools.com. (n.d.). <https://www.w3schools.com/html/>
- 4) *Social website from scratch - Web design Beginner Tutorial | PHP, HTML, CSS, MYSQL Database*. (n.d.). YouTube.
<https://www.youtube.com/playlist?list=PLY3j36HMSHNWaKUC73RJlwi6oU-WTpTPM>